

JeanNassar

Senior Python Backend Engineer – Robotics & Data Systems

+32 499 41 28 47
jean@jnassar.com
jnassar.com
masasin
in masasin
34M, Canadian
Master of Engineering



Summary Senior Backend Engineer with 10+ years of experience building and stabilizing resilient software systems. Proven expertise in architecting scalable Python applications (FastAPI, SQLAlchemy) and implementing comprehensive testing strategies. Multidisciplinary background in robotics (ROS, C++) and data pipelines.

Work experience

- Sep 2023– Jun 2025 **Senior Software Developer, Nipro Digital Technologies Europe, Brugge, Belgium**
- Led end-to-end design and implementation of critical backend features for a complex medical data application (Ruby on Rails, PostgreSQL).
 - Architected the foundational backend for a new analytics dashboard, processing production data with dbt and Cube.js.
 - Delivered a from-scratch OIDC/LDAP authentication integration, meeting enterprise security and interoperability requirements.
- Jan 2021– Aug 2023 **Senior Software Developer, Adimian, Brussels, Belgium**
- Architected and developed resilient backend systems and data processing pipelines using Python, FastAPI, Pydantic, and SQLAlchemy.
 - Took end-to-end ownership of projects, from initial design and development through to deployment, monitoring, and maintenance.
 - Implemented comprehensive test suites using pytest, ensuring high code quality and maintainability for all deliverables.
 - Mentored colleagues in Python and backend development best practices, improving team skills and onboarding new members.
- Dec 2019– Aug 2020 **Senior Python Developer, Yields.io, Brussels, Belgium**
- Systematically refactored and stabilized a complex Python codebase, introducing thousands of tests to greatly improve stability and reliability.
 - Engineered a Docker-based integration testing environment to replace unreliable mocks, ensuring accurate, real-world API validation.
 - Led the migration and automated testing of client code and artifacts across platform versions.
- Mar–Oct 2019 **Industrial Automation Engineer, Kapernikov, Brussels, Belgium**
- Developed a conveyor belt monitoring system using Python 3 and ROS.
 - Used a laser profiler and camera to create a 3D representation of the environment for real-time object detection.
 - Fixed bugs in C++ drivers and created a standalone ROS node for camera communication.
- Feb–Dec 2018 **Data Scientist/Python Developer, Sentiance, Antwerp, Belgium**
- Led the migration of the company's core codebase from Python 2 to 3.
 - Refactored monolithic functionality into modular, testable components.
 - Utilized PySpark to parallelize data processing workloads and improve efficiency.

Projects

- SPIRIT Masters thesis project: third-person view for a monocular UAV
- Created a system that superimposed a CGI version of a drone on top of an actual image taken by its FPV camera earlier
 - Designed, conducted, and analyzed user studies which showed a large improvement in many metrics, even with 2 Hz transmission
- Yozakura Software development lead for a teleoperated rescue robot
- Developed client-server system for robot control: programmed Raspberry Pi and mbed chips, including servo and sensor drivers
- Open Source Projects Two packages on PyPI, many repositories on github, and contributions to other projects such as numpy, ROS, and more
- latexipy (129 stars) for exporting Matplotlib-based plots into native $\text{\LaTeX}$, this résumé (118 stars), AoC 2021 and 2022, and more

Education

- 2014–2017 **MEng, Mechanical Engineering**, *Kyoto University*, Mechatronics Laboratory, Kyoto, Japan
- Thesis: A UAV Teleoperation System Using Subimposed Past Images
 - Tutored Python and software development best practices
- 2008–2013 **BASc, Honours Mechatronics Engineering**, *University of Waterloo*, Waterloo, ON, Canada
- Completed six co-op internships in three countries
 - Cofounded the Engineering Ambassador program
 - Directed the UW EngSoc Mental Health Awareness directorship

Technical skills

- Backend Python (incl. pytest, FastAPI), SQLAlchemy, PostgreSQL, Redis
- DevOps Docker, AWS, CI/CD, Git
- Robotics C++, ROS, RPi, ESP8266, Arduino, mbed, MicroPython, PCB design
- Software Scientific Python Stack, Jupyter, Ruby (Rails), Javascript (Vue), LLMs
- CAD OnShape, Fusion, Solidworks, Inventor, OpenSCAD
- OSes Linux, macOS, Windows

Languages

- Native English, French, Lebanese
- Fluent Japanese, Arabic
- Intermediate Dutch, Spanish